Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Cisco Certified Network Professional (CCNP) for ENCOR Training

TGS Ref No: TGS-2024044052

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 6.0

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
5.0	1 April 2024	Previous release — 5-day CCNP ENCOR lesson plan built on the legacy master trainer slides.	Dr Alfred Ang
6.0	7 September 2026	Full rebuild against the Cisco ENCOR 350-401 v1.1 exam blueprint. Six topics realigned to the six exam domains with their published weightings; 35 hands-on Packet Tracer labs replacing the legacy discovery exercises; schedule rebuilt to 9:30am–6:30pm with 8 instructional hours per day; slide references now generated directly from the deck.	Dr Alfred Ang

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

TABLE OF CONTENTS

Course Information **4**
Learning Outcomes **4**
Skills Framework Mapping **4**
Instructional Methods **5**
Assessment **5**
Course Schedule **5**
Day 1 — Enterprise Architecture — design, switching paths, campus LAN and high availability 5
Day 2 — Virtualization Technologies — hypervisors, VRF, GRE/IPsec, LISP and VXLAN 6
Day 3 — Network Infrastructure — Layer 2, EIGRP/OSPF/BGP and IP services 7
Day 4 — Network Infrastructure (wireless), Network Assurance and Network Security 8
Day 5 — Network Automation, revision and final assessment 9
Lab Reference — Aligned to the ENCOR 350-401 Exam Domains **10**
Lab Index with Slide References **10**
Resources Required **13**

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Course Information

Course Title	Cisco Certified Network Professional (CCNP) for ENCOR Training
WSQ Course Reference	TGS-2024044052
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	5 days · 8 training hours per day (40 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks counted within training time)
Mode	Instructor-led, with hands-on Cisco Packet Tracer configuration and troubleshooting labs
Certification Prepared For	Cisco 350-401 ENCOR v1.1 — Implementing Cisco Enterprise Network Core Technologies
Skills Framework	TSC: Network Configuration (ICT-DIT-3009-1.1)
Trainer	Dr Alfred Ang
Trainer Slides	Cisco Certified Network Professional (CCNP) for ENCOR Training-v6.0.pptx (211 slides)

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Translate organizational networking needs into technical configurations by selecting appropriate network components, considering their functions and interdependencies.
- LO2: Identify and choose key network components for configuration, utilizing techniques in server and device implementation aligned with technical requirements.
- LO3: Implement a variety of servers and devices effectively, following the organizational network blueprint to enhance network performance and reliability.
- LO4: Manage network access, ensuring compliance with organizational guidelines and specifications to maintain secure and efficient user connectivity.
- LO5: Select and deploy network management tools effectively, optimizing network performance and user experience in line with organizational needs.
- LO6: Conduct regular network reporting and administration, employing assessment tools to ensure network stability and alignment with organizational objectives.

Skills Framework Mapping

This course is mapped to the Skills Framework TSC Network Configuration (ICT-DIT-3009-1.1).

Knowledge (K)	Ability (A)
K1: Key network components – their functions and interdependencies	A1: Translate the organisational requirements to technical requirements
K2: Techniques in implementation and configuration of servers, devices and other network components	A2: Identify the key network components to be configured to meet the technical requirements
K3: Features of network management tools	A3: Implement multiple servers and devices in line with the organisation's network blueprint

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

K4: Configuration of network servers and devices	A4: Manage user network access in accordance to organisational guidelines and specifications
K5: Network access management	A5: Identify suitable network management tools to be used
K6: Network tracking and assessment tools	A6: Implement procedures to ensure regular network reporting, administration and assessment of stability

Instructional Methods

- Lecture and trainer-led explanation of each exam domain's concepts, supported by diagrams, comparison tables and worked examples in the trainer slides.
- Didactic questioning throughout each topic to check understanding and surface misconceptions before the labs.
- Demonstration of each configuration on Cisco IOS before learners attempt it.
- Practical hands-on labs in Cisco Packet Tracer — 35 labs, one or more for every exam objective in the blueprint.
- Guided troubleshooting exercises in which learners diagnose a deliberately broken configuration rather than building from blank.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on Cisco configuration and verification tasks, 1 hour, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- The final assessment is conducted on Day 5 from 4:30 pm, following the Briefing for Assessment and the Assessment digital attendance.
- Learners are assessed as Competent (C) or Not Yet Competent (NYC) on each instrument. An appeal process is available.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Each training day totals 8 instructional hours. The 1-hour lunch break is not counted as instructional time; tea breaks are counted within it. Slide references are to the trainer slide deck listed in Course Information.

Day 1 — Enterprise Architecture — design, switching paths, campus LAN and high availability

Time	Duration	Topic / Activity
9:30–10:15	45 min	Welcome, trainer and learner introductions, ground rules, learning outcomes, course outline, Skills Framework alignment and mandatory digital attendance (AM) (Slides

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

		2–21)
10:15–11:15	60 min	**Core concepts — the ENCOR 350-401 blueprint, the three planes, switching mechanisms, the forwarding tables and the campus design decision (Slides 22–46)**
11:15–11:30	15 min	Tea break
11:30–13:00	90 min	**Topic 1 — Enterprise Architecture: hierarchical design, high availability, wireless design principles, SD-WAN and SD-Access planes, QoS components and policy (Slides 47–52)**
13:00–14:00	60 min	Lunch break
14:00–15:45	105 min	Digital attendance (PM). Hands-on: Lab 1: Investigate the CAM and TCAM Tables; Lab 2: Analyse Cisco Express Forwarding; Lab 3: Compare Two-Tier and Three-Tier Campus Designs (Slides 53–61)
15:45–16:00	15 min	Tea break
16:00–18:15	135 min	Hands-on: Lab 4: Interpret Wired and Wireless QoS Configurations; Lab 5: Explore SD-WAN and SD-Access Architecture (Slides 62–67)
18:15–18:30	15 min	Day 1 recap, Q&A and consolidation of Topic 1

Total instructional time: 480 minutes (8 hours).

Day 2 — Virtualization Technologies — hypervisors, VRF, GRE/IPsec, LISP and VXLAN

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and mandatory digital attendance (AM)
9:45–11:15	90 min	**Topic 2 — Virtualization Technologies: Type 1 and Type 2 hypervisors, virtual machines, virtual switching, and the concepts behind VRF, GRE, IPsec, LISP and VXLAN (Slides 69–75)**
11:15–11:30	15 min	Tea break
11:30–13:00	90 min	Hands-on: Lab 6: Configure and Verify VRF-Lite (Slides 76–78)
13:00–14:00	60 min	Lunch break
14:00–15:45	105 min	Digital attendance (PM). Hands-on: Lab 7: Configure and Verify a GRE Tunnel; Lab 8: Configure and Verify a Site-to-Site IPsec VPN (Slides 79–

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

		84)
15:45–16:00	15 min	Tea break
16:00–18:15	135 min	Consolidation of the data path virtualization objectives. Hands-on: Lab 9: Explore LISP and VXLAN Network Virtualization (Slides 85–87)
18:15–18:30	15 min	Day 2 recap, Q&A and consolidation of Topic 2

Total instructional time: 480 minutes (8 hours).

Day 3 — Network Infrastructure — Layer 2, EIGRP/OSPF/BGP and IP services

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 2 recap and mandatory digital attendance (AM)
9:45–11:15	90 min	**Topic 3 — Network Infrastructure (Layer 2): 802.1Q trunking and DTP, EtherChannel with PAgP and LACP, STP, RSTP, MST and the Spanning Tree protection features (Slides 89–98)**
11:15–11:30	15 min	Tea break
11:30–13:00	90 min	Hands-on: Lab 10: Troubleshoot 802.1Q Trunking and DTP; Lab 11: Troubleshoot Static and Dynamic EtherChannels (Slides 99–104)
13:00–14:00	60 min	Lunch break
14:00–15:45	105 min	Digital attendance (PM). Hands-on: Lab 12: Configure RSTP and Tune the Spanning Tree; Lab 13: Configure Multiple Spanning Tree (MST); Lab 14: Harden the Spanning Tree with PortFast, BPDU Guard and Root Guard (Slides 105–113)
15:45–16:00	15 min	Tea break
16:00–18:15	135 min	Topic 3 (Layer 3): EIGRP versus OSPF path selection, multiarea OSPFv2 and OSPFv3, eBGP and policy-based routing. Hands-on: Lab 15: Compare EIGRP and OSPF Path Selection; Lab 16: Configure Multiarea OSPFv2 with Summarization and Filtering; Lab 17: Configure Multiarea OSPFv3 for IPv6; Lab 18: Configure and Verify eBGP (Slides 114–125)
18:15–18:30	15 min	Day 3 recap, Q&A and consolidation of the Layer 2 and Layer 3 objectives

Total instructional time: 480 minutes (8 hours).

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Day 4 — Network Infrastructure (wireless), Network Assurance and Network Security

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 3 recap and mandatory digital attendance (AM)
9:45–11:15	90 min	Topic 3 (wireless): Layer 1 concepts, AP modes and antennas, CAPWAP discovery and join, Layer 2 and Layer 3 roaming, and wireless segmentation. Hands-on: Lab 19: Configure a Basic WLAN and WPA2-Enterprise on the WLC; Lab 20: Troubleshoot WLAN and Wireless Client Connectivity (Slides 126–131)
11:15–11:30	15 min	Tea break
11:30–13:00	90 min	Topic 3 (IP services): NTP and PTP, NAT and PAT, HSRP, VRRP and GLBP, and multicast with RPF, PIM and IGMP. Hands-on: Lab 21: Configure NTP, NAT/PAT and HSRP (Slides 132–134)
13:00–14:00	60 min	Lunch break
14:00–15:45	105 min	Digital attendance (PM). Topic 4 — Network Assurance: diagnostics, Flexible NetFlow, SPAN/RSPAN/ERSPAN, IP SLA, DNA Center workflows and NETCONF/RESTCONF (Slides 136–141). Hands-on: Lab 22: Diagnose Network Problems with Ping, Traceroute and Conditional Debug; Lab 23: Configure Syslog, SNMP and Flexible NetFlow; Lab 24: Configure SPAN, RSPAN and IP SLA; Lab 25: Configure NETCONF and RESTCONF and Explore DNA Center Workflows (Slides 142–153)
15:45–16:00	15 min	Tea break
16:00–18:15	135 min	Topic 5 — Network Security: device access control, AAA, ACLs, CoPP, REST API security, wireless security and security design (Slides 155–162). Hands-on: Lab 26: Configure Secure Device Access with SSH and Strong Passwords; Lab 27: Implement Local and Server-Based AAA; Lab 28: Configure Extended ACLs to Mitigate Attacks; Lab 29: Configure Control Plane Policing; Lab 30: Analyse

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

		Wireless Security, REST API Security and Security Design (Slides 163–177)
18:15–18:30	15 min	Day 4 recap, Q&A and consolidation of Topics 4 and 5

Total instructional time: 480 minutes (8 hours).

Day 5 — Network Automation, revision and final assessment

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 4 recap and mandatory digital attendance (AM)
9:45–11:15	90 min	Topic 6 — Network Automation: interpreting Python, constructing valid JSON, YANG data modelling (Slides 179–185). Hands-on: Lab 31: Interpret Python Components and Scripts; Lab 32: Construct and Validate JSON-Encoded Files (Slides 186–191)
11:15–11:30	15 min	Tea break
11:30–13:00	90 min	Hands-on: Lab 33: Explore YANG Models and Call the DNA Center API; Lab 34: Construct an EEM Applet to Automate Response (Slides 192–197)
13:00–14:00	60 min	Lunch break
14:00–15:15	75 min	Digital attendance (PM). Hands-on: Lab 35: Compare Agent and Agentless Orchestration Tools (Slides 198–200)
15:15–16:00	45 min	**Course revision across all six exam domains, exam-technique guidance and the practice exam at exams.tertiaryinfotech.com (Slides 208–211)**
16:00–16:15	15 min	Course feedback and mandatory TRAQOM survey
16:15–16:30	15 min	**Briefing for Assessment and Assessment digital attendance**
16:30–17:30	60 min	**Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book**
17:30–18:30	60 min	**Practical Performance (PP) — hands-on Cisco configuration and verification tasks, 1 hour, open book**

Total instructional time: 480 minutes (8 hours).

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Lab Reference — Aligned to the ENCOR 350-401 Exam Domains

Topic / Exam domain	Weighting	TSC	Labs
Topic 01: Enterprise Architecture (ENCOR domain 1.0 Architecture)	15%	K1, A1	Labs 1–5
Topic 02: Virtualization Technologies (ENCOR domain 2.0 Virtualization)	10%	K2, A2	Labs 6–9
Topic 03: Network Infrastructure (ENCOR domain 3.0 Infrastructure)	30%	K3, A3	Labs 10–21
Topic 04: Network Assurance (ENCOR domain 4.0 Network Assurance)	10%	K5, A4	Labs 22–25
Topic 05: Network Security (ENCOR domain 5.0 Security)	20%	K4, A6	Labs 26–30
Topic 06: Network Automation (ENCOR domain 6.0 Automation)	15%	K6, A5	Labs 31–35

Lab Index with Slide References

Lab	Title	Exam objective	Slide
1	Investigate the CAM and TCAM Tables	Describe hardware switching mechanisms such as CAM, TCAM, FIB, RIB and adjacency tables (ENCOR 1.6).	53
2	Analyse Cisco Express Forwarding	Describe and verify CEF, the FIB and the adjacency table, and contrast them with process and fast switching (ENCOR 1.6).	56
3	Compare Two-Tier and Three-Tier Campus Designs	Explain enterprise design principles including 2-tier, 3-tier, fabric and cloud designs (ENCOR 1.1.a).	59
4	Interpret Wired and Wireless QoS Configurations	Interpret wired and wireless QoS configurations, including QoS components and QoS policy (ENCOR 1.5).	62
5	Explore SD-WAN and SD-Access Architecture	Explain the working principles of the Cisco SD-WAN and SD-Access solutions and their control and data plane elements (ENCOR 1.3, 1.4).	65
6	Configure and Verify	Configure and verify data path virtualization	76

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

	VRF-Lite	technologies — VRF (ENCOR 2.2.a).	
7	Configure and Verify a GRE Tunnel	Configure and verify data path virtualization technologies — GRE tunneling (ENCOR 2.2.b).	79
8	Configure and Verify a Site-to-Site IPsec VPN	Configure and verify data path virtualization technologies — IPsec tunneling (ENCOR 2.2.b).	82
9	Explore LISP and VXLAN Network Virtualization	Describe network virtualization concepts — LISP and VXLAN (ENCOR 2.3.a, 2.3.b).	85
10	Troubleshoot 802.1Q Trunking and DTP	Troubleshoot static and dynamic 802.1q trunking protocols (ENCOR 3.1.a).	99
11	Troubleshoot Static and Dynamic EtherChannels	Troubleshoot static and dynamic EtherChannels — PAgP and LACP (ENCOR 3.1.b).	102
12	Configure RSTP and Tune the Spanning Tree	Configure and verify common Spanning Tree Protocols — RSTP — and Spanning Tree enhancements (ENCOR 3.1.c).	105
13	Configure Multiple Spanning Tree (MST)	Configure and verify common Spanning Tree Protocols — MST (ENCOR 3.1.c).	108
14	Harden the Spanning Tree with PortFast, BPDU Guard and Root Guard	Configure and verify Spanning Tree enhancements such as root guard and BPDU guard (ENCOR 3.1.c).	111
15	Compare EIGRP and OSPF Path Selection	Compare routing concepts of EIGRP and OSPF — advanced distance vector versus link state, load balancing, path selection, metrics and area types (ENCOR 3.2.a).	114
16	Configure Multiarea OSPFv2 with Summarization and Filtering	Configure simple OSPFv2 environments including multiple normal areas, summarization and filtering (ENCOR 3.2.b).	117
17	Configure Multiarea OSPFv3 for IPv6	Configure simple OSPFv3 environments including multiple normal areas (ENCOR 3.2.b).	120
18	Configure and Verify eBGP	Configure and verify eBGP between directly connected neighbours, including best path selection and neighbour relationships (ENCOR 3.2.c).	123
19	Configure a Basic WLAN and WPA2-Enterprise on the WLC	Describe AP discovery and join, wireless segmentation with groups, profiles and tags,	126

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

		and configure wireless security features (ENCOR 3.3.c, 3.3.f, 5.4).	
20	Troubleshoot WLAN and Wireless Client Connectivity	Troubleshoot WLAN configuration and wireless client connectivity issues using GUI only (ENCOR 3.3.e).	129
21	Configure NTP, NAT/PAT and HSRP	Interpret NTP configurations, configure NAT/PAT, and configure first hop redundancy protocols such as HSRP and VRRP (ENCOR 3.4.a, 3.4.b, 3.4.c).	132
22	Diagnose Network Problems with Ping, Traceroute and Conditional Debug	Diagnose network problems using tools such as debugs, conditional debugs, traceroute, ping, SNMP and syslog (ENCOR 4.1).	142
23	Configure Syslog, SNMP and Flexible NetFlow	Configure and verify Flexible NetFlow, and use SNMP and syslog for diagnosis (ENCOR 4.1, 4.2).	145
24	Configure SPAN, RSPAN and IP SLA	Configure SPAN/RSPAN/ERSPAN and configure and verify IPSLA (ENCOR 4.3, 4.4).	148
25	Configure NETCONF and RESTCONF and Explore DNA Center Workflows	Configure and verify NETCONF and RESTCONF, and describe Cisco DNA Center workflows (ENCOR 4.5, 4.6).	151
26	Configure Secure Device Access with SSH and Strong Passwords	Configure and verify device access control — lines and local user authentication (ENCOR 5.1.a).	163
27	Implement Local and Server-Based AAA	Configure and verify device access control — authentication and authorization using AAA (ENCOR 5.1.b).	166
28	Configure Extended ACLs to Mitigate Attacks	Configure and verify infrastructure security features — ACLs (ENCOR 5.2.a).	169
29	Configure Control Plane Policing	Configure and verify infrastructure security features — CoPP (ENCOR 5.2.b).	172
30	Analyse Wireless Security, REST API Security and Security Design	Describe REST API security, configure wireless security features, and describe the components of network security design (ENCOR 5.3, 5.4, 5.5).	175
31	Interpret Python Components and Scripts	Interpret basic Python components and scripts (ENCOR 6.1).	186

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

32	Construct and Validate JSON-Encoded Files	Construct valid JSON-encoded files (ENCOR 6.2).	189
33	Explore YANG Models and Call the DNA Center API	Describe the principles and benefits of YANG, describe APIs for Cisco DNA Center and vManage, and interpret REST API response codes (ENCOR 6.3, 6.4, 6.5).	192
34	Construct an EEM Applet to Automate Response	Construct an EEM applet to automate configuration, troubleshooting or data collection (ENCOR 6.6).	195
35	Compare Agent and Agentless Orchestration Tools	Compare agent versus agentless orchestration tools such as Chef, Puppet, Ansible and SaltStack (ENCOR 6.7).	198

Resources Required

- Cisco Packet Tracer 8.2 or later installed on each learner workstation (free from the Cisco Networking Academy).
- The lab pack for this course — one folder per lab containing the topology file, the addressing table and the mock data set.
- Python 3.10 or later with the requests library, for the Topic 6 automation labs.
- Internet access for the REST API labs and the practice exam at exams.tertiaryinfotech.com.
- Projector or large display for trainer demonstrations, and a whiteboard for topology sketches.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg